

29 The atomic spacing in a thin metal film is of the order of magnitude 10^{-10} m .

For the diffraction of an electron beam, what order of magnitude of velocity is required?

A 10^4 m s^{-1}

B 10^5 m s^{-1}

C 10^6 m s^{-1}

D 10^7 m s^{-1}

30 The transitions of electrons between three consecutive energy levels in an atom give rise